Quiz #1 (Part I - Extended): Computer Systems & Data Representation

Total Marks: 22.00

Architecture, Machine Arithmetic & Floating-Point Systems

Name: **Student ID:** **Score:** / 22.00

Instructions: Answer all twelve questions. Pay close attention to the question-type tags and technique hints. Calculators are strictly prohibited.

Question 1

Mark 2.00 out of 2.00

[MULTIPLE CHOICE - SELECT ONE]

Technique / Concept Hint: Interactive command line execution: syntax evaluation of standard output commands with operator and operand arguments.

In early interactive command-line computer environments, how were arithmetic operations executed and evaluated directly from the command prompt?

- a. By using the output command `print` followed by two operands and one operator (e.g., `print 8 + 4`)
- b. By invoking an iterative loop script containing memory register declarations
- c. By manually toggling CPU accumulator switches without invoking system output routines
- d. By compiling a separate binary program file directly through a batch job

Question 2

Mark 2.00 out of 2.00

[MULTIPLE CHOICE - SELECT ONE]

Technique / Concept Hint: Multi-core processor nomenclature: distinguishing physical execution units (cores) on a single die from logical threads or sockets.

What is an octa-core CPU?

- a. A CPU that executes eight virtual threads on a single core via simultaneous multithreading
- b. A physical CPU chip that contains eight independent processing cores (execution units)
- c. A motherboard architecture designed with eight separate physical CPU sockets
- d. A CPU equipped with eight discrete levels of Level 1 (L1) data cache

Question 3

Mark 2.00 out of 2.00

[SELECT ALL THAT APPLY]

Technique / Concept Hint: IEEE 754 structural bitfield mapping: sign bit conventions, exponent bias handling, and the hidden/implicit significand bit.

Which of the following statements are **TRUE** regarding the bitfield conventions of the IEEE 754 floating-point standard (both single-precision 32-bit and double-precision 64-bit)? (Select all options that apply.)

- [ ] a. The first digit on the far left (the most significant bit) designates the sign of the number in both formats.
- [ ] b. A sign bit of `0` represents a positive value, while `1` represents a negative value.
- [ ] c. The exponent field is stored in raw two's complement without any bias added.
- [ ] d. In normalized numbers, an implicit leading bit of `1` precedes the fractional mantissa bits.

Question 4

Mark 2.00 out of 2.00

[FILL IN THE BLANK - CALCULATION]

Technique / Concept Hint: Real decimal to binary conversion: successive division by 2 for the whole integer part, and successive multiplication by 2 tracking whole-number carries for the fractional part.

What is the exact binary representation of the decimal number **22.375**?

Binary equivalent: _____ $_2$

Question 5

Mark 1.00 out of 1.00

[SELECT ALL THAT APPLY]

Technique / Concept Hint: Unsigned binary positional weighting: evaluating binary parity (even vs. odd) using the 2^0 position (LSB) and bitfield bounds (MSB).

Consider the decimal number **78** represented as an 8-bit unsigned binary integer. Which of the following statements apply? (Select all that apply.)

- [ ] a. The least significant bit (LSB, bit 0) is `0` .
- [ ] b. The least significant bit (LSB, bit 0) is `1` .
- [ ] c. The most significant bit (MSB, bit 7) is `0` .
- [ ] d. Exactly four bits in this 8-bit pattern are set to `1` .

Question 6

Mark 2.00 out of 2.00

[FILL IN THE BLANK - CALCULATION]

Technique / Concept Hint: IEEE 754 float decoding: determine sign from bit 31, calculate real exponent as (Biased Exponent – 127), restore the implicit 1 to mantissa (1.F), and perform binary shifts.

Information about a 32-bit single-precision IEEE 754 floating-point number is demonstrated below:

```
Sign = 0
Exponent = 10000011
Mantissa = 00101000000000000000000
```

What is the decimal value (in base-10) represented by this 32-bit floating-point pattern?

Decimal value:

Question 7

Mark 2.00 out of 2.00

[MULTIPLE CHOICE - SELECT ONE]

Technique / Concept Hint: IEEE 754 normalization mechanics: implicit bit optimization and precision maximization.

Why does the normalized IEEE 754 floating-point representation omit the leading 1 (storing only the fractional part in the mantissa field)?

- a. Because the hardware ALU cannot process bits of value 1 when reading the significand
- b. To gain an extra bit of precision for free, knowing all normalized binary numbers start with `1.`
- c. To ensure all negative numbers can be represented using exact two's complement format
- d. To allow denormalized numbers to share identical hardware paths without checking flags

Question 8

Mark 2.00 out of 2.00

[SELECT ALL THAT APPLY]

Technique / Concept Hint: Biased exponent properties: comparison efficiency, formula $2^{k-1} - 1$, and range distinctions between single and double precision.

Which of the following statements are **TRUE** regarding biased exponent representation in IEEE 754? (Select all that apply.)

- [ ] a. In single precision (32-bit), the bias added to the actual exponent is `127` .
- [ ] b. In double precision (64-bit), the bias added to the actual exponent is `1023` .
- [ ] c. A biased exponent allows floating-point numbers to be compared for magnitude using integer comparison hardware without first decoding sign bits.
- [ ] d. An 8-bit exponent containing all `1` s (`11111111`) represents normal numbers with very large positive powers.

Question 9

Mark 2.00 out of 2.00

[FILL IN THE BLANK - CALCULATION]

Technique / Concept Hint: Negative powers of 2 encoding: normalization shift count determination and exponent biasing formula ($\text{Biased } E = \text{Actual } E + 127$).

Consider encoding the decimal number **−0.125** ($-{}^{1}/_{8}$) into IEEE 754 single-precision (32-bit) format:

(a) Decimal value of the biased exponent: _____

(b) 8-bit binary pattern of the exponent: _____ 2

Question 10

Mark 2.00 out of 2.00

[FILL IN THE BLANK - CALCULATION]

Technique / Concept Hint: Fixed-point normalization and mantissa extraction: shift radix point to obtain $1.F$, then extract the 23-bit fractional mantissa field padded with trailing zeros.

When converting the decimal number **−25.25** into IEEE 754 single-precision (32-bit) format, what is the exact **23-bit stored mantissa**?

Stored 23-bit mantissa: _____

Question 11

Mark 2.00 out of 2.00

[SELECT ALL THAT APPLY]

Technique / Concept Hint: IEEE 754 special values: analyzing reserved bit patterns for Zero, Infinity, and NaN (Not-a-Number).

Which of the following statements correctly describe the IEEE 754 single-precision representation of special numerical cases? (Select all that apply.)

- [ ] a. Positive and negative zeros are distinguished by having all zero bits in both exponent and mantissa, differing only by the sign bit.
- [ ] b. An exponent of all `1` s with an all- `0` mantissa represents positive or negative Infinity (±∞).
- [ ] c. An exponent of all `1` s accompanied by a non-zero mantissa represents NaN (Not a Number).
- [ ] d. Division by zero in IEEE 754 always immediately triggers a fatal processor crash rather than producing a bit pattern.

Question 12

Mark 1.00 out of 1.00

[FILL IN THE BLANK - CALCULATION]

Technique / Concept Hint: Two's complement integer encoding: invert all bits of the positive magnitude and add 1, then verify the most significant bit (MSB, bit 7).

What is the **Most Significant Bit (MSB)** of the decimal integer **−42** when represented as an 8-bit signed two's complement integer?

MSB (Bit 7): _____

Instructor Answer Key & Detailed Marking Guide

Maximum Marks: 22.00

Quiz #1 (Part I - Extended) - Comprehensive Solutions

Question 1 (2.00 Marks) — Multiple Choice

Correct Answer: a

Applied Concept: Early command-line shells directly evaluated expressions provided to the standard output routine using two operands and a single operator (e.g., `print 8 + 4`)[cite: 1].

Question 2 (2.00 Marks) — Multiple Choice

Correct Answer: b

Applied Concept: Core CPU terminology: "octa-" indicates eight physical processor execution units embedded onto a single chip package[cite: 1].

Question 3 (2.00 Marks) — Select All That Apply

Correct Answers: a, b, d

Applied Concept: IEEE 754 bitfields:

- **a is TRUE:** Bit 31 (single) or Bit 63 (double) is universally the sign bit[cite: 1].
- **b is TRUE:** `0` signifies positive, and 1 signifies negative.
- **c is FALSE:** Exponent fields use biased (excess) notation, not two's complement[cite: 2].
- **d is TRUE:** Normalized mantissas omit the leading `1.` to conserve memory[cite: 2].

Question 4 (2.00 Marks) — Fill in the Blank

Correct Answer: 10110.011

Step-by-step arithmetic:

1. **Integer (22):** 22 / 2 = 11 (rem 0); 11 / 2 = 5 (rem 1); 5 / 2 = 2 (rem 1); 2 / 2 = 1 (rem 0); 1 / 2 = 0 (rem 1) → 10110_2
2. **Fraction (0.375):** 0.375 × 2 = 0.750 (carry 0); 0.750 × 2 = 1.500 (carry 1); 0.500 × 2 = 1.000 (carry 1) → $.011_2$
3. **Combined:** 10110.011_2.

Question 5 (1.00 Mark) — Select All That Apply

Correct Answers: a, c, d

Applied Technique: Decimal 78 in 8-bit unsigned binary is 01001110_2 (64 + 8 + 4 + 2).

- The LSB (bit 0) is `0` (even parity), making **a** true and **b** false[cite: 1].
- The MSB (bit 7) is `0`, making **c** true.
- The bits set to 1 are at positions 6, 3, 2, and 1 (four 1s total), making **d** true.

Question 6 (2.00 Marks) — Fill in the Blank

Correct Answer: 18.5

Step-by-step decoding:

1. **Sign:** Bit is `0` → Positive (+).
2. **Biased Exponent:** 10000011_2 = 128 + 2 + 1 = 131[cite: 2].
3. **Unbiased Exponent:** 131 – 127 = 4[cite: 2].
4. **Mantissa:** Stored = `0010100...` → Significand = 1.00101_2[cite: 2].
5. **Binary Shift:** $+1.00101_2 \times 2^4 = 10010.1_2$.
6. **Decimal Conversion:** 16 + 2 + 0.5 = **18.5**.

Question 7 (2.00 Marks) — Multiple Choice

Correct Answer: b

Applied Concept: Since every normalized binary number has a leading non-zero digit (which must be 1 in base 2), this bit is implicit and omitted from storage, providing 24 bits of precision from a 23-bit mantissa field[cite: 2].

Question 8 (2.00 Marks) — Select All That Apply

Correct Answers: a, b, c

Applied Concept:

• **a & b are TRUE:** The bias for single precision is $2^{8-1}-1 = 127$; for double precision it is $2^{11-1}-1 = 1023$[cite: 2].

• **c is TRUE:** Biasing turns negative exponents into smaller positive integers, allowing standard integer comparison logic to compare exponent orders without checking signs.

• **d is FALSE:** An exponent field of all 1s (255) is strictly reserved for Infinity and NaN[cite: 2].

Question 9 (2.00 Marks) — Fill in the Blank

Correct Answers: (a) 124, (b) 01111100

Step-by-step calculation:

1. 0.125 = $1/8 = 2^{-3}$.
2. In normalized format: $-1.0_2 \times 2^{-3}$ → Actual exponent $E = -3$.
3. Biased Exponent = –3 + 127 = **124**[cite: 2].
4. 124 converted to 8-bit unsigned binary: 64 + 32 + 16 + 8 + 4 = **01111100_2**.

Question 10 (2.00 Marks) — Fill in the Blank

Correct Answer: 10010100000000000000000

Step-by-step extraction:

1. 25 in binary = 11001_2; 0.25 in binary = $.01_2$ → 11001.01_2.
2. Normalized: $1.100101_2 \times 2^4$[cite: 2].
3. Strip the leading `1.` to leave the fractional mantissa: `100101`[cite: 2].
4. Pad with trailing zeros to complete the 23-bit field: **10010100000000000000000**[cite: 2].

Question 11 (2.00 Marks) — Select All That Apply

Correct Answers: a, b, c

Applied Concept:

• **a is TRUE:** IEEE 754 provides signed zero (+0 and –0) with all exponent and mantissa bits set to zero.

• **b is TRUE:** Exponent = 255 and Mantissa = 0 encodes ±∞.
• **c is TRUE:** Exponent = 255 and Mantissa ≠ 0 encodes NaN.
• **d is FALSE:** IEEE 754 specifies silent or signaled production of ±∞ or NaN on invalid arithmetic, rather than hardware system crashes.

Question 12 (1.00 Mark) — Fill in the Blank

Correct Answer: 1

Step-by-step arithmetic:

1. Positive 42 in 8-bit binary: 00101010_2.
2. Invert bits (one's complement): 11010101_2.
3. Add 1 (two's complement): 11010110_2.
4. The Most Significant Bit (MSB, bit 7) is **1** (indicating a negative value).
